

AI Marketing Campaign Management System

Backend API & Database Technical Specification Document

1. System Overview & Technology Stack

This document details the backend architecture, database schemas, and RESTful API endpoints for the **AI Marketing Campaign Management System**. The backend has been modernized to a high-performance Python ecosystem using **FastAPI**, **SQLAlchemy ORM**, and **MySQL** (with SQLite fallback). It seamlessly handles marketing campaign workflows, social media post creation, profile management, notifications, audit logging, and proxying AI generation requests to **n8n automation webhooks**.

Component	Technology / Specification	Role / Purpose
Web Framework	FastAPI (Python 3.12)	Async REST API server with automatic OpenAPI docs
Database ORM	SQLAlchemy 2.0	Object-Relational Mapping & Database Abstraction
RDBMS	MySQL 8.0+ / PyMySQL	Primary Relational Database Storage
Data Validation	Pydantic v2	Request/Response validation & <code>_id</code> alias mapping
AI Automation	n8n Webhook Cloud Proxy	AI content generation pipeline integration

2. Database Schemas (Models)

The system consists of 5 core relational database tables designed with SQLAlchemy. To ensure backward compatibility with the React frontend (which consumes MongoDB ``_id`` fields), the Pydantic response layer dynamically computes ``_id`` as a string representation of the integer ``id`` primary key.

2.1 Campaign Table (campaigns)

Field Name	Data Type	Constraints / Default	Description
id / <code>_id</code>	Integer / String	Primary Key, Auto-increment	Unique Identifier
name	VARCHAR(255)	NOT NULL	Campaign Title
objective	VARCHAR(255)	NOT NULL	Lead Gen, Brand Awareness, etc.
status	VARCHAR(50)	Default: 'Active'	Active, Paused, Completed
budget	Float	NOT NULL	Budget allocated in USD
roi	Float	Default: 0.0	Return on Investment (%)
startDate	VARCHAR(50)	NOT NULL	Campaign start date (YYYY-MM-DD)
endDate	VARCHAR(50)	NOT NULL	Campaign end date (YYYY-MM-DD)
channels	JSON / Text	Default: []	Array of target channels

2.2 Post Table (posts)

Field Name	Data Type	Constraints / Default	Description
id / <code>_id</code>	Integer / String	Primary Key, Auto-increment	Post Identifier

title	VARCHAR(255)	NOT NULL	Post Title
platform	VARCHAR(100)	NOT NULL	Instagram, LinkedIn, Twitter, etc.
content	TEXT	NOT NULL	Social media post copy
imageUrl	TEXT	Default: "	Image asset URL
publishedAt	DATETIME	Default: CURRENT_TIMESTAMP	Publication timestamp

2.3 Profile Table (profiles)

Field Name	Data Type	Constraints / Default	Description
id / _id	Integer / String	Primary Key, Auto-increment	Profile Identifier
name	VARCHAR(255)	NOT NULL	User Full Name
email	VARCHAR(255)	NOT NULL	User Email Address
company	VARCHAR(255)	NOT NULL	Organization Name
phone	VARCHAR(100)	Default: "	Contact Number
industry	VARCHAR(255)	Default: "	Business Sector
avatar	TEXT	Default: "	Avatar Image URL or Base64

2.4 Notification Table (notifications)

Field Name	Data Type	Constraints / Default	Description
id / _id	Integer / String	Primary Key, Auto-increment	Notification Identifier
title	VARCHAR(255)	NOT NULL	Alert Header
text	TEXT	NOT NULL	Notification Body
type	VARCHAR(50)	Default: 'info'	'info', 'success', 'error'
unread	Boolean	Default: True	Read Status Flag

2.5 Audit History Table (history_logs)

Field Name	Data Type	Constraints / Default	Description
id / _id	Integer / String	Primary Key, Auto-increment	Log Entry Identifier
action	VARCHAR(255)	NOT NULL	Action Description
category	VARCHAR(100)	NOT NULL	'campaign', 'post', 'profile', etc.
details	TEXT	NOT NULL	Detailed event context

3. API Specification Catalog

The backend exposes 15 RESTful endpoints. All responses are formatted in strict JSON.

Method	Endpoint Path	Request Body	Description
GET	/api/campaigns	None	Retrieve list of all active/past campaigns
POST	/api/campaigns	CampaignCreate JSON	Create a new campaign entry
PUT	/api/campaigns/{id}	CampaignUpdate JSON	Update campaign details by ID
DELETE	/api/campaigns/{id}	None	Delete a campaign by ID
GET	/api/posts	None	Retrieve social media posts
POST	/api/posts	PostCreate JSON	Create social media post
DELETE	/api/posts/{id}	None	Delete social media post by ID
GET	/api/profile	None	Fetch user profile details
PUT	/api/profile	ProfileUpdate JSON	Update user profile settings
GET	/api/notifications	None	Fetch top 10 recent notifications
POST	/api/notifications	NotificationCreate JSON	Add new notification
PUT	/api/notifications/mark-read	None	Mark all notifications as read
PUT	/api/notifications/{id}/read	None	Mark specific notification as read
GET	/api/history	None	Retrieve system activity audit logs
POST	/api/history	HistoryCreate JSON	Add audit log entry

POST	/api/generate-ai-post	AIPostRequest JSON	Proxy single AI copy gen to n8n webhook
POST	/api/generate-ai-batch	AIBatchRequest JSON	Proxy batch multi-channel AI copy to n8n

4. How to Run & Verify

1. Start FastAPI Server:

cd backend && python main.py

2. Interactive API Documentation (Swagger UI):

Navigate to <http://localhost:5000/docs> in any browser to test API endpoints interactively.

3. Environment Variables (.env):

Set DATABASE_URL=mysql+pymysql://root:password@localhost:3306/ai_marketing in backend/.env.